Presentation Layer

Menu_HR
+userName: String
+service: Service
+ printMenu() : void
+ getAnswer() : void
- printShiftOptions() : void
- printDayOptions() : void

Main
+ main(String[] args) : void
+ handleEmployeeLogin(Scanner scanner) : void
+ handleHRLogin(Scanner scanner) : void

Menu_User
+userName: String
+service: Service
+ printMenu() : void
+ getAnswer() : void

Service Layer

Service
+userService: UserService
+shiftService: ShiftService
+ Service()
+ login(String userName, String password) : boolean
+ changePassword(String userName, String newPassword) : boolean
+ pickAvailability(String userName, String input) : boolean
+ showMyAvailability(String userName) : void
+ changeAvailability(String userName, String oldShift, String newShift) : boolean
+ updateBankAccount(String userName, String bank, String branch, String account) : boolean
+ showMyPersonalDetails(String userName) : void
+ logout(String userName) : boolean
+ isHRManager(String userName) : boolean
+ createRole(String roleId, String roleName, String description) : boolean
+ addRoleToEmployee(String employeeUserName, String roleId) : boolean
+ hireEmployee(String employeeUserName, String employeeId, String password, String branchId, String hourlySalary, String employmentType, String bankNumber, String bankBranchNumber, String bankAccountNumber) : boolean
+ fireEmployee(String employeeUserName) : boolean
+ changeEmployeeSalary(String employeeUserName, String newSalary) : boolean
+ showAllEmployeesAndRoles() : void
+ showAllRoles() : void
+ showRoleOptions() : void
+ showNextWeekShifts() : void
+ publishNextWeek() : boolean
+ publishNextWeekRequirements(String deadlineOption, String customDate) : boolean
+ setAsCurrentWeek() : boolean
+ changeShiftRequirement(String branchId, String date, String shiftType, String roleId, String amount) : boolean
+ markHolidayDay(String branchId, String shiftNumber, String week) : boolean
+ changeShiftRequirement(String branchId, String date, String shiftType, String roleId, String amount, String week) : boolean
+ changeShiftRequirementByShiftNumber(String branchId, String shiftNumber, String roleId, String amount, String week) : boolean
+ showAvailableEmployeesForShiftByRole(String branchId, String date, String shiftType, String roleId) : void
+ showAvailableEmployeesForShiftByRole(String branchId, String shiftNumber, String roleId) : void
+ assignEmployee(String employeeUserName, String branchId, String date, String shiftType, String roleId) : boolean
+ assignEmployeeByShiftNumber(String employeeUserName, String branchId, String shiftNumber, String roleId) : boolean
+ removeEmployeeFromShiftByShiftNumber(String employeeUserName, String branchId, String shiftNumber) : boolean
+ showNextWeekStatus() : void
+ showCurrentWeekShifts() : void

UserService
-employeeController: EmployeeController
+ UserService()
+ changePassword(String userName, String newPassword) : boolean
+ pickAvailability(String userName, String input) : boolean
+ showMyAvailability(String userName) : void
+ changeAvailability(String userName, String oldInput, String newInput) : boolean
+ updateBankAccount(String userName, String bankStr, String branchStr, String accountStr) : boolean
+ showMyPersonalDetails(String userName) : void
+ logout(String userName) : boolean
+ login(String userName, String password) : boolean
+ isHRManager(String userName) : boolean
+ createRole(String roleIdStr, String roleName, String description) : boolean
- parseShifts(String input) : ArrayList
- isValidShift(int shift) : boolean
- shiftToText(int shift) : String
+ addRoleToEmployee(String employeeUserName, String roleInput) : boolean
+ hireEmployee(String employeeUserName, String employeeId, String password, String branchIdStr, String hourlySalaryStr, String employmentTypeStr, String bankNumberStr, String bankBranchNumberStr, String bankAccountNumberStr) : boolean
+ fireEmployee(String employeeUserName) : boolean
+ changeEmployeeSalary(String employeeUserName, String newSalaryStr) : boolean
+ showAllEmployeesAndRoles() : void
+ showAllRoles() : void
+ showRoleOptions() : void
- printAvailabilityStatus(Employee employee) : void

ShiftService
-shiftController: ShiftController
-employeeController: EmployeeController
+ ShiftService()
+ syncCurrentWeekIfNeeded() : void
+ publishNextWeek() : boolean
+ publishNextWeekRequirements(String deadlineOption, String customDate) : boolean
+ setAsCurrentWeek() : boolean
+ changeShiftRequirement(String branchIdStr, String dateStr, String shiftTypeStr, String roleIdStr, String amountStr) : boolean
+ markHolidayDay(String branchIdStr, String shiftNumberStr, String weekStr) : boolean
+ showNextWeekStatus() : void
+ changeShiftRequirementByShiftNumber(String branchIdStr, String shiftNumberStr, String roleInput, String amountStr, String weekStr) : boolean
+ changeShiftRequirement(String branchIdStr, String dateStr, String shiftTypeStr, String roleInput, String amountStr, String weekStr) : boolean
+ showAvailableEmployeesForShiftByRole(String branchIdStr, String dateStr, String shiftTypeStr, String roleInput) : void
+ showAvailableEmployeesForShiftByRole(String branchIdStr, String shiftNumberStr, String roleInput) : void
+ assignEmployee(String employeeUserName, String branchIdStr, String dateStr, String shiftTypeStr, String roleInput) : boolean
+ assignEmployeeByShiftNumber(String employeeUserName, String branchIdStr, String shiftNumberStr, String roleInput) : boolean
+ removeEmployeeFromShiftByShiftNumber(String employeeUserName, String branchIdStr, String shiftNumberStr) : boolean
+ showShiftRequirements(String branchIdStr, String dateStr, String shiftTypeStr) : void
+ showShiftRequirements(String branchIdStr, String dateStr, String shiftTypeStr, String weekStr) : void
+ showShiftRequirementsByShiftNumber(String branchIdStr, String shiftNumberStr, String weekStr) : void
+ showCurrentWeekShifts() : void
+ showNextWeekShifts() : void
+ showAllShiftsHistory() : void
+ showShiftsHistoryByBranch(String branchIdStr) : void
- parseAvailabilityDeadline(String deadlineOption, String customDate) : LocalDateTime

Domain Layer

<<singleton>> EmployeeController
-employees: ArrayList
-firedEmployees: ArrayList
-branches: ArrayList
-roles: ArrayList
- EmployeeController()
+ getInstance() : EmployeeController
+ addEmployee(Employee employee) : boolean
+ getEmployee(String userName) : Employee
+ employeeExists(String userName) : boolean
+ addRole(Role role) : boolean
+ getRole(int roleID) : Role
+ getRoleByName(String roleName) : Role
+ getRoleByIdOrName(String roleInput) : Role
+ roleExists(int roleID) : boolean
+ addRoleToEmployee(String userName, int roleID) : boolean
+ employeeHasRole(String userName, int roleID) : boolean
+ removeEmployee(String userName) : boolean
+ changeEmployeeSalary(String userName, int newSalary) : boolean
+ employeesAndRolesToString() : String
+ rolesAndEmployeeCountToString() : String
+ rolesListToString() : String
+ addBranch(int branchID) : boolean
+ branchExists(int branchID) : boolean
+ getEmployees() : ArrayList
+ getBranches() : ArrayList
+ getFiredEmployees() : ArrayList
+ getRoles() : ArrayList

<<singleton>> ShiftController
-employeeController: EmployeeController
-shiftOrganizers: Map
- ShiftController()
+ getInstance() : ShiftController
+ publishNextWeek() : boolean
+ publishNextWeekRequirements(LocalDateTime deadline) : boolean
+ nextWeekMissingRequirementsToString() : String
+ syncCurrentWeekIfNeeded() : void
+ setAsCurrentWeek() : boolean
+ isNextWeekLocked(int branchId) : boolean
+ hasPublishedNextWeekRequirements(int branchId) : boolean
+ canEmployeesUpdateAvailability(int branchId) : boolean
+ getAvailabilityDeadline(int branchId) : LocalDateTime
+ nextWeekStatusToString() : String
+ nextWeekPublishSummaryToString() : String
+ changeShiftRequirement(int branchId, LocalDate date, Shift.ShiftType shiftType, int roleId, int amount) : boolean
+ markHolidayDay(int branchId, int shiftIndex, ShiftWeek week) : boolean
+ changeShiftRequirement(int branchId, int shiftIndex, int roleId, int amount, ShiftWeek week) : boolean
+ changeShiftRequirement(int branchId, LocalDate date, Shift.ShiftType shiftType, int roleId, int amount, ShiftWeek week) : boolean
+ setDefaultRequirementForAllShifts(int branchId, Role role, int amount) : boolean
+ availableEmployeesForShiftByRoleToString(int branchId, LocalDate date, Shift.ShiftType shiftType, int roleId) : String
+ availableEmployeesForShiftByRoleToString(int branchId, int shiftIndex, int roleId) : String
+ assignEmployee(String userName, int branchId, LocalDate date, Shift.ShiftType shiftType, int roleId) : boolean
+ assignEmployee(String userName, int branchId, int shiftIndex, int roleId) : boolean
+ removeEmployeeFromShift(String userName, int branchId, int shiftIndex) : boolean
+ currentWeekShiftsToString() : String
+ nextWeekShiftsToString() : String
+ allShiftsHistoryToString() : String
+ shiftsHistoryByBranchToString(int branchId) : String
+ shiftRequirementsToString(int branchId, LocalDate date, Shift.ShiftType shiftType) : String
+ shiftRequirementsToString(int branchId, int shiftIndex, ShiftWeek week) : String
+ shiftRequirementsToString(int branchId, LocalDate date, Shift.ShiftType shiftType, ShiftWeek week) : String
- getOrCreateOrganizer(int branchId) : ShiftOrganizer
- setShiftRequirement(Shift shift, Role role, int amount) : void
- findNextWeekShiftIndex(ShiftOrganizer organizer, LocalDate date, Shift.ShiftType shiftType) : int
- findShiftIndex(ShiftOrganizer organizer, LocalDate date, Shift.ShiftType shiftType, ShiftWeek week) : int
- weekToText(ShiftWeek week) : String

Role
-roleName: String
-description: String
-roleID: int
+ Role(int roleID, String roleName, String description)
+ getRoleID() : int
+ getRoleName() : String
+ getDescription() : String
+ setDescription(String description) : void
+ equals(Object o) : boolean
+ hashCode() : int
+ toString() : String

Employee
-userName: String
-password: String
-id: String
-isLoggedIn: boolean
-bankAccount: BankAccount
-branchID: int
-vacationDays: int
-roles: ArrayList
-hourlySalary: int
-startContract: LocalDate
-endContract: LocalDate
-availabilities: ArrayList
-employmentType: EmploymentType
+ Employee(int BID, String userName, BankAccount bankAccount, int hourlySalary, String enterPassword, EmploymentType employmentType, String id)
+ getUserName() : String
+ setUserName(String userName) : void
+ getPassword() : String
+ setPassword(String password) : void
+ getBranchID() : int
+ setBranchID(int branchID) : void
+ getRoles() : ArrayList
+ setRoles(ArrayList roles) : void
+ getId() : String
+ getVacationDays() : int
+ setVacationDays(int vacationDays) : void
+ useVacationDays(int days) : boolean
+ addVacationDays(int days) : void
+ getHourlySalary() : int
+ setHourlySalary(int hourlySalary) : void
+ getStartContract() : LocalDate
+ setStartContract(LocalDate startContract) : void
+ getEndContract() : LocalDate
+ setEndContract(LocalDate endContract) : void
+ setIsLoggedIn(boolean isLoggedIn) : void
+ getIsLoggedIn() : boolean
+ getBankAccount() : BankAccount
+ setBankAccount(BankAccount bankAccount) : void
+ getEmploymentType() : EmploymentType
+ setEmploymentType(EmploymentType employmentType) : void
+ checkPassword(String input) : boolean
+ addRole(Role role) : void
+ removeRole(Role role) : void
+ getAvailabilities() : ArrayList
+ setAvailabilities(ArrayList availabilities) : void
+ addAvailability(int shift) : void
+ hasAvailability(int shift) : boolean
+ replaceAvailability(int oldShift, int newShift) : void
- rolesToString() : String
- availabilitiesToString() : String
+ toString() : String

Shift
-shiftId: int
-branchId: int
-date: LocalDate
-shiftType: ShiftType
-closedDay: boolean
-requirements: Map
-employeeRoleAssignments: Map
+ Shift(int shiftId, int branchId, LocalDate date, ShiftType shiftType)
+ addRequirement(Role role, int minimumCount) : void
+ removeRequirement(Role role) : void
+ clearRequirements() : void
+ addBackRequirement(Role role) : void
+ getRequirements() : Map
+ assignEmployee(Employee employee, Role role) : void
+ fulfillRequirement(Role role) : boolean
+ removeEmployee(Employee employee) : boolean
+ getRoleOf(Employee employee) : Role
+ isAssigned(Employee employee) : boolean
+ requiresRole(Role role) : boolean
+ getRequiredCountForRole(Role role) : int
+ isClosedDay() : boolean
+ setClosedDay(boolean closedDay) : void
+ getAssignedEmployees() : List
+ getEmployeeRoleAssignments() : Map
+ getAssignmentCount() : int
- hasShiftManager() : boolean
+ getShiftId() : int
+ getBranchId() : int
+ getDate() : LocalDate
+ getShiftType() : ShiftType
+ equals(Object obj) : boolean
+ hashCode() : int
+ toString() : String

ShiftOrganizer
-branchId: int
-currentWeekShifts: Shift[]
-nextWeekShifts: Shift[]
-availableEmployeesPerShiftByRole: Map>
-weekStartDate: LocalDate
-availabilityChangesAllowed: boolean
-isPublished: boolean
-requirementsPublishedAt: LocalDateTime
-availabilityDeadline: LocalDateTime
-shiftsHistory: List
+ ShiftOrganizer(int branchId, Shift[] currentWeek, Shift[] nextWeek)
+ ShiftOrganizer(int branchId)
+ getBranchId() : int
+ getWeekStartDate() : LocalDate
+ isAvailabilityChangesAllowed() : boolean
+ isPublished() : boolean
+ hasPublishedNextWeekRequirements() : boolean
+ canEmployeesUpdateAvailability() : boolean
+ getAvailabilityDeadline() : LocalDateTime
+ getCurrentWeekShifts() : Shift[]
+ getNextWeekShifts() : Shift[]
+ getCurrentWeekShift(int shiftIndex) : Shift
+ getNextWeekShift(int shiftIndex) : Shift
+ publishNextWeek() : boolean
+ publishNextWeekRequirements(LocalDateTime deadline) : boolean
+ setNextWeekAsCurrentWeek() : boolean
+ shouldAutoAdvanceToNextWeek() : boolean
+ changeShiftRequirement(int shiftIndex, Role role, int minimumCount) : boolean
+ changeShiftRequirement(int shiftIndex, Role role, int minimumCount, boolean forNextWeek) : boolean
+ markHolidayDay(int shiftIndex, boolean forNextWeek) : boolean
+ updateAvailableEmployees(List employees) : void
+ getAvailableEmployeesForShiftByRole(int shiftIndex, Role role) : List
+ assignEmployee(int shiftIndex, Employee employee, Role role) : boolean
+ removeEmployeeFromShift(int shiftIndex, Employee employee) : boolean
+ showCurrentWeekShifts() : String
+ showNextWeekShifts() : String
+ showAllShiftsHistory() : String
+ showShiftsHistoryByBranch(int branchId) : String
+ getShiftsHistory() : List
+ areAllNextWeekRequirementsSatisfied() : boolean
+ nextWeekMissingRequirementsToString() : String
+ nextWeekStatusToString() : String
+ nextWeekPublishSummaryToString() : String
- addAvailableEmployee(Shift shift, Role role, Employee employee) : void
- employeeHasRole(Employee employee, Role role) : boolean
- isEmployeeAssignedOnSameDate(Employee employee, LocalDate date) : boolean
- prepareWeek(Shift[] shifts, LocalDate defaultStartDate) : Shift[]
- createWeekShifts(LocalDate startDate) : Shift[]
- getStartOfCurrentWeek() : LocalDate
- validateShiftIndex(int shiftIndex) : void
- shiftsToString(Shift[] shifts) : String
- formatShiftName(Shift shift) : String

BankAccount
-bankNumber: int
-bankBranchNumber: int
-bankAccountNumber: int
+ BankAccount(int bankNumber, int bankBranchNumber, int bankAccountNumber)
+ getBankNumber() : int
+ setBankNumber(int bankNumber) : void
+ getBankBranchNumber() : int
+ setBankBranchNumber(int bankBranchNumber) : void
+ getBankAccountNumber() : int
+ setBankAccountNumber(int bankAccountNumber) : void
+ toString() : String

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